KAN Technology Overview

Kolmogorov-Arnold Networks for nonlinear MRI classification

Project: Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Technical overview | Research and project-evaluation reference

1. What is a KAN?

Kolmogorov-Arnold Networks (KANs) are neural network architectures motivated by the Kolmogorov-Arnold representation theorem. In the original KAN formulation, learnable univariate functions are placed on edges, in contrast to conventional multilayer perceptrons that typically use fixed activations at nodes and learned scalar weights on edges.

Reference: Liu et al., “KAN: Kolmogorov-Arnold Networks” (2024).

2. How a KAN layer works

A conventional dense layer computes weighted sums followed by an activation function. A KAN-style layer instead learns functions phi_ij(x_i) for individual input dimensions and combines their outputs. Different implementations approximate these functions using B-splines, radial basis functions or other parameterizations.

Simplified relation: y_j = sum_i phi_ij(x_i). In a spline KAN, phi_ij is often represented by a learnable spline basis. An RBF-KAN uses radial basis functions and is a variant, not the same as the original spline implementation.

3. Why use KAN in your brain tumor project?

- Replace the conventional dense classification head with a KAN-style nonlinear mapping.
- Investigate whether learned nonlinear features help distinguish glioma, meningioma, pituitary and no-tumor images.
- Compare parameter count, macro-F1, calibration, inference latency and training stability against an ordinary MLP head.
- Do not claim that KAN automatically guarantees interpretability or improved diagnostic accuracy.

4. Integration in the hybrid pipeline

Module	Role
EfficientNet-B0	Local feature extraction from MRI
Vision Mamba	Global-context representation of spatial tokens
Pooling	Convert token sequence to a compact embedding
KAN head	Learn nonlinear mapping from embedding to four logits
Temperature scaling	Calibrate output confidence on validation data

5. Variants and implementation decisions

Variant	Notes
Original spline KAN	Uses learnable spline functions; may require careful grid and numerical tuning
RBF-KAN	Uses radial basis functions to approximate univariate nonlinear maps
FastKAN-style variants	Aim to reduce computational cost, but differ from original KAN
MLP baseline	Essential control to test whether KAN adds value

6. Research ablation experiments

- Experiment A: EfficientNet-B0 + MLP classification head.
- Experiment B: EfficientNet-B0 + KAN classification head.
- Experiment C: EfficientNet-B0 + Vision Mamba + MLP head.
- Experiment D: EfficientNet-B0 + Vision Mamba + KAN head.
- Run matched splits and seeds; compare macro-F1, per-client recall, calibration, GPU time and bytes per federated round.

7. Practical limitations

- KAN layers can be computationally costly at high input dimensionality; use a compact embedding and a controlled hidden width.
- Different KAN implementations are not interchangeable. State exactly whether you used spline KAN, RBF-KAN or another variant.
- Complexity can cause overfitting on small MRI datasets. Use appropriate regularization, held-out validation and repeated seeds.
- Any explanation of tumor location should come from a validated spatial method such as Grad-CAM, not from KAN outputs alone.

8. Viva questions and references

- Why KAN instead of a dense layer? To test learnable nonlinear edge functions as an alternative classifier.
- Is KAN always better? No; it must outperform or otherwise improve on a controlled MLP baseline.
- What does RBF-KAN mean? A KAN-inspired architecture using radial basis function expansions.

Liu, Z., et al. (2024). KAN: Kolmogorov-Arnold Networks. https://arxiv.org/abs/2404.19756

Li, Z. (2024). Kolmogorov-Arnold Networks are Radial Basis Function Networks. https://arxiv.org/abs/2405.06721